



CEBU INSTITUTE OF TECHNOLOGY
U N I V E R S I T Y

IT342-G1

System Integration and Architecture

System Design Document (SDD)

Project Title: AssetFlow

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Version: 0.6

Date: February 28, 2026

Status: Final

REVISION HISTORY TABLE

Version	Date	Author	Changes Made	Status
0.1	February 21, 2026	Salonga, Andre D.	Initial draft	Draft
0.2	February 28, 2026	Salonga, Andre D.	Added API specifications	Review
0.3	February 28, 2026	Salonga, Andre D.	Updated database design	Review
0.4	February 28, 2026	Salonga, Andre D.	Added UI/UX designs	Review
0.5	February 28, 2026	Salonga, Andre D.	Incorporated feedback	Revised
0.6	February 28, 2026	Salonga, Andre D.	Final review and corrections	Final
1	[Date]	[Your Name]	Baseline version for development	Approved

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EXECUTIVE SUMMARY

1.1 Project Overview

AssetFlow is a centralized web and mobile-based tracking system designed to streamline the management of organizational equipment. Developed for the Systems Integration and Architecture 1 course, the platform provides real-time visibility into asset availability and ensures user accountability. It utilizes a modern three-tier architecture, consisting of a Spring Boot backend, a React web frontend, and an Android mobile client.

1.2 Objectives

1. Provide a searchable inventory for laboratory keepers and officers to manage equipment status.
2. Implement robust identity management using BCrypt hashing and JWT for session persistence.
3. Restrict administrative functions (adding assets, marking returns) to authorized officers while allowing students to request items.
4. Incorporate external APIs, automated email notifications (SMTP), and secure file uploads for asset documentation.

1.3 Scope

Included Features:

- User registration/login
- JWT-based session management
- Searchable inventory dashboard
- Borrow request transactions
- Administrative asset management

Excluded Features:

- Asset depreciation tracking
- Automated late fee payments
- Multi-organization support

1.0 INTRODUCTION

1.1 Purpose

This document serves as the technical blueprint for AssetFlow, translating functional requirements into a detailed design specification. It outlines the architectural decisions, database schema, and API contracts necessary to ensure a secure and scalable implementation.

2.0 FUNCTIONAL REQUIREMENTS SPECIFICATION

2.1 Project Overview

Project Name: AssetFlow

Domain: Inventory and Resource Management

Primary Users: Organization Officers (Admins), Laboratory Keepers, and Student Members

Problem Statement: Organizations and school laboratories often struggle with manual or disorganized methods for tracking equipment, leading to lost assets, lack of availability data, and zero user accountability.

Solution: A centralized asset management system utilizing a client-server architecture (React, Android, and Spring Boot) that provides real-time visibility into item status and maintains a digital audit trail of all borrowing and returning transactions.

2.2 Core User Journeys

Journey 1: Account Creation (New Member)

1. User navigates to the registration page.
2. User enters required details including Email and Password.
3. The system validates the input and hashes the password using BCrypt.
4. The system stores the user record and confirms successful registration.

Journey 2: Secure Access (Registered Officer/Member)

1. User enters valid credentials on the login form.
2. The system authenticates the user and generates a JWT.
3. User is redirected to the Inventory Dashboard where they can see item availability.

4. The system displays specific tools based on the user's role (e.g., "Add Asset" for Officers).

Journey 3: Session Termination

1. The authenticated user clicks the "Logout" button.
2. The system removes the JWT from the local application state and invalidates the session.
3. The user is redirected to the login screen and restricted from making further borrow requests.

Journey 4: Equipment Borrowing (Student Member)

1. User logs into the AssetFlow mobile or web app.
2. User searches the Inventory Dashboard for an "Available" asset.
3. User submits a Borrow Request.
4. The system updates the asset status to "Borrowed," logs the user's ID and timestamp, and sends a confirmation email via SMTP.

Journey 5: Asset Return and Recovery (Laboratory Keeper)

1. The Laboratory Keeper (Admin) logs into the system.
2. The Keeper identifies the asset being returned in the Admin Tools dashboard.
3. The Keeper marks the item as "Returned".
4. The system updates the status back to "Available" and archives the transaction in the history.

2.3 Feature List (MoSCoW)

MUST HAVE

1. User Registration/Login with JWT.
2. Full CRUD for Assets (Major Entity).
3. Borrow/Return transaction logic.
4. Role-Based Access Control (Admin vs. Member).
5. Email Notifications for borrow confirmations.

SHOULD HAVE

1. File Upload for asset photos or condition reports.
2. External API integration (e.g., a barcode/QR lookup or weather API for outdoor gear)

COULD HAVE

1. Real-Time dashboard updates using WebSockets.
2. Google OAuth login for student accounts.

WON'T HAVE

1. Asset Depreciation Calculation
2. Automated Late Fee Payments
3. Multi-Organization Support
4. Physical Hardware Integration

2.4 Detailed Feature Specifications

Feature: User Identity & Security

- **Screens:** Registration, Login, Profile
- **Fields:** First Name, Last Name, Email Address, Password, Confirm Password
- **Validation:** Email format, password matching, field completion, and uniqueness.
- **API Endpoints:** POST /api/auth/register, POST /api/auth/login, POST /api/auth/logout, GET /api/auth/me.
- **Security:** JWT tokens, password hashing with BCrypt, and Google OAuth integration.
- **Notifications:** Automated "Welcome Email" sent upon successful registration via SMTP.

Feature: Asset Inventory Dashboard

- **Screens:** Inventory Listing, Asset Detail.
- **Display:** Grid/List layout showing asset images, status (Available, Borrowed, Maintenance), and category.
- **Search/Filter:** Search by asset name or filter by availability status.

- **External Integration:** Meaningful consumption of a Public API (e.g., Weather API to display outdoor conditions for field equipment).
- **API Endpoints:** GET /api/assets, GET /api/assets/{id}, GET /api/assets/search

Feature: Transactional Module (Borrowing & Returns)

- **Screens:** Asset Detail (Borrow button), My Transactions History.
- **Functions:** Select "Available" item, submit borrow request, log borrower ID and timestamp.
- **Persistence:** Database storage tracking one-to-many relationship between User and Transactions.
- **API Endpoints:** POST /api/transactions/borrow, GET /api/transactions/history.
- **Notifications:** Automated "Borrow Confirmation" receipt sent to user via SMTP.

Feature: File & Asset Management

- **Screens:** Add Asset Form, Edit Asset Page.
- **Functions:** Upload asset images or PDF manuals, link files to database records.
- **Storage:** Files stored on server or cloud, accessible for user view/download.
- **API Endpoints:** POST /api/assets/upload, GET /api/assets/download/{fileId}

Feature: Administrative Tools

- **Screens:** Asset Management Dashboard, User Role Management.
- **Functions:** Full CRUD (Create, Read, Update, Delete) on Assets; Mark items as "Returned" or "Under Maintenance".
- **Access Control:** Admin/Laboratory Keeper role required for all management endpoints.
- **API Endpoints:** POST /api/assets, PUT /api/assets/{id}, DELETE /api/assets/{id}, PATCH /api/transactions/{id}/return.

2.5 Acceptance Criteria

AC-1: Successful User Registration & Notification

Given I am a guest user on the registration page.

When I enter a unique email, Student ID, and matching passwords.

Then my account should be stored with a BCrypt hashed password.

And the system must automatically send a "Welcome Email" via SMTP to my registered address.

AC-2: Secure Login & Google OAuth Integration

Given I am a registered member or officer.

When I log in with valid credentials or via Google OAuth.

Then the system must generate a unique JWT for my session.

And I should be redirected to the Inventory Dashboard.

AC-3: Asset Borrowing Transaction

Given I am an authenticated student on the Asset Detail page.

When I select an "Available" asset and click the "Borrow" button.

Then the asset status must update to "Borrowed" in the database.

And the system must log my User ID and a timestamp, then send a borrow confirmation email.

AC-4: Role-Based Access Control (Admin Tools)

Given I am authenticated with a "Student Member" role.

When I attempt to access the "Add New Asset" form or the /api/assets POST endpoint.

Then the system must deny access with a 403 Forbidden error.

AC-5: Asset Return & File Management (Admin)

Given I am authenticated as a "Laboratory Keeper" or "Officer".

When I mark a borrowed item as "Returned" and upload a photo of its current condition.

Then the file must be stored on the server and linked to the asset record.

And the asset status must return to "Available".

3.0 NON-FUNCTIONAL REQUIREMENTS

3.1 Performance Requirements

- **API Response Time:** The Spring Boot backend must process 95% of API requests (e.g., fetching inventory or submitting borrow requests) within 2 seconds.
- **Mobile App Responsiveness:** The Android application must remain responsive during API calls, using asynchronous Retrofit calls to ensure the UI does not freeze.
- **Database Efficiency:** Search queries for assets must complete within 500ms, even as the inventory grows.
- **Concurrency:** The system must support at least 50 concurrent users (students and officers) without performance degradation.

3.2 Security Requirements

- **Authentication:** All protected routes in both the React web app and Android mobile app must require a valid JWT (JSON Web Token).
- **Data Integrity:** All user passwords must be hashed using BCrypt before being stored in the database.
- **Authorization:** The system must implement Role-Based Access Control (RBAC), ensuring only "Admin" or "Laboratory Keeper" roles can access CRUD operations for assets.
- **Secure Communication:** All data exchange between the clients (Web/Mobile) and the Spring Boot API must occur over HTTPS.
- **OAuth Security:** After a successful Google OAuth login, the system must generate and return a custom JWT to maintain session security.

3.3 Compatibility Requirements

- **Mobile OS:** The Android application must be compatible with API Level 34 (Android 14) and developed using XML-based UI layouts (Jetpack Compose is strictly prohibited).
- **Web Browsers:** The React web application must be fully functional on the latest versions of Google Chrome, Mozilla Firefox, and Safari.
- **Backend Environment:** The server must run on Java 17+ to support the Spring Boot 3.x framework.

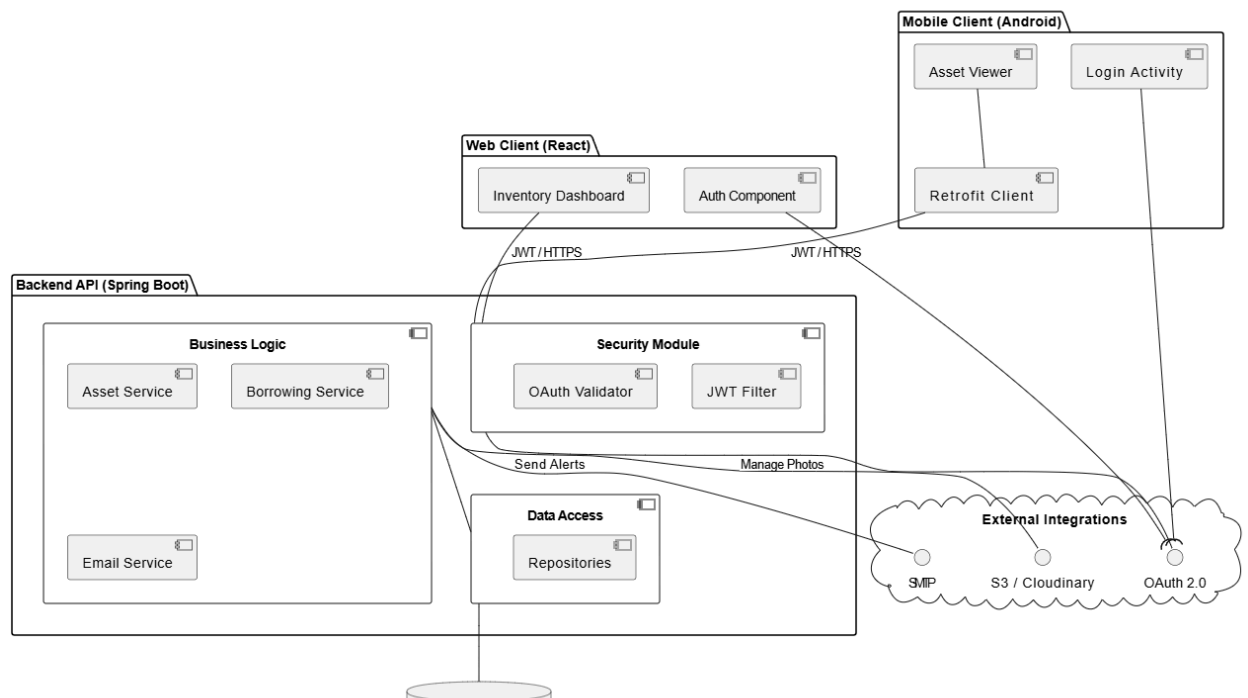
- Database: The system must be compatible with relational database management systems like PostgreSQL or MySQL.

3.4 Usability Requirements

- Role-Aware UI: The user interface must dynamically adjust based on the user's role (e.g., hiding the "Add Asset" button for student members).
- File Management: The system must provide a clear interface for administrators to upload asset photos and for users to view/download them.
- Notification Clarity: Automated emails (SMTP) for registration and borrow confirmations must be delivered within 60 seconds of the triggering event.
- Error Handling: The system must provide meaningful error messages for failed logins, unauthorized access (403 Forbidden), or invalid file uploads.
- Mobile Navigation: The Android app should follow standard navigation patterns (e.g., bottom navigation or drawer) with touch targets of at least 44x44px.

4.0 SYSTEM ARCHITECTURE

4.1 Component Diagram



Technology Stack:

- **Backend:** Java 17, Spring Boot 3.x, Spring Security (JWT), Spring Data JPA
- **Database:** PostgreSQL 14+ / MySQL 8.0
- **Web Frontend:** React 18, JavaScript, Tailwind CSS, Axios
- **Mobile:** Kotlin, XML-based Layouts, Retrofit, Room
- **Build Tools:** Maven (Backend), npm/yarn (Web), Gradle (Android)
- **Deployment:** Render/Railway (Backend), Vercel/Netlify (Web), APK (Mobile)

5.0 API CONTRACT & COMMUNICATION

5.1 API Standards

- **Base URL:** <https://api.assetflow.cit.edu/api/v1>
- **Format:** JSON for all requests and responses.
- **Authentication:** Bearer token (JWT) must be included in the Authorization header for all protected endpoints.
- **Response Structure:**

json

```
{  
  "success": boolean,  
  "data": object|null,  
  "error": {  
    "code": string,  
    "message": string,  
    "details": object|null  
  },  
  "timestamp": "ISO-8601 string"  
}
```

5.2 Endpoint Specifications

Authentication Endpoints

POST /auth/google:

Body: { "idToken": "string" }

Response: { "user": { "id", "email", "name", "role" }, "token", "refreshToken" }

POST /auth/logout:

Headers: Authorization: Bearer {token}

Response: { "message": "Logged out successfully" }

Asset Endpoints

GET /assets:

Query: ?page=1&limit=20&search=keyword&category=tools

Response: { "assets": [...], "pagination": { "page", "limit", "total" } }

GET /assets/{id}:

Response: { "asset": { "id", "name", "serialNumber", "status", "imageUrl", "category" } }

POST /assets (Admin/Officer only):

Body: { "name", "serialNumber", "category", "imageFile" }

Response: { "asset": { ... } }

Borrowing Endpoints

POST /borrow/request:

Body: { "assetId": "long", "expectedReturnDate": "string" }

Response: { "message": "Request submitted", "transaction": { ... } }

PATCH /borrow/status/{transactionId} (Admin/Officer only):

Body: { "status": "APPROVED|REJECTED|RETURNED" }

Response: { "message": "Status updated", "transaction": { ... } }

5.3 Error Handling

HTTP Status Codes

- 200 OK – Successful request
- 201 Created – Resource (e.g., Asset or User) created
- 400 Bad Request – Invalid input or missing fields
- 401 Unauthorized – Authentication required/JWT failed
- 403 Forbidden – Insufficient permissions (e.g., Student accessing Admin features)
- 404 Not Found – Asset or Transaction doesn't exist
- 409 Conflict – Duplicate resource (e.g., Serial Number already exists)
- 500 Internal Server Error – Server-side error or Database connection failure

Error Code Examples

json

```
{  
  "success": false,  
  "data": null,  
  "error": {  
    "code": "AUTH-002",  
    "message": "Token expired",  
    "details": "The provided JWT has expired. Please log in again via Google."  
  },  
  "timestamp": "2026-03-01T23:10:00Z"  
}
```

```
{  
  "success": false,  
  "data": null,  
  "error": {  
    "code": "VALID-001",  
    "message": "Validation failed",  
    "details": {  
      "assetName": "Asset name is required",  
      "serialNumber": "Serial number must be unique"  
    }  
  },  
  "timestamp": "2026-03-01T23:10:00Z"  
}
```

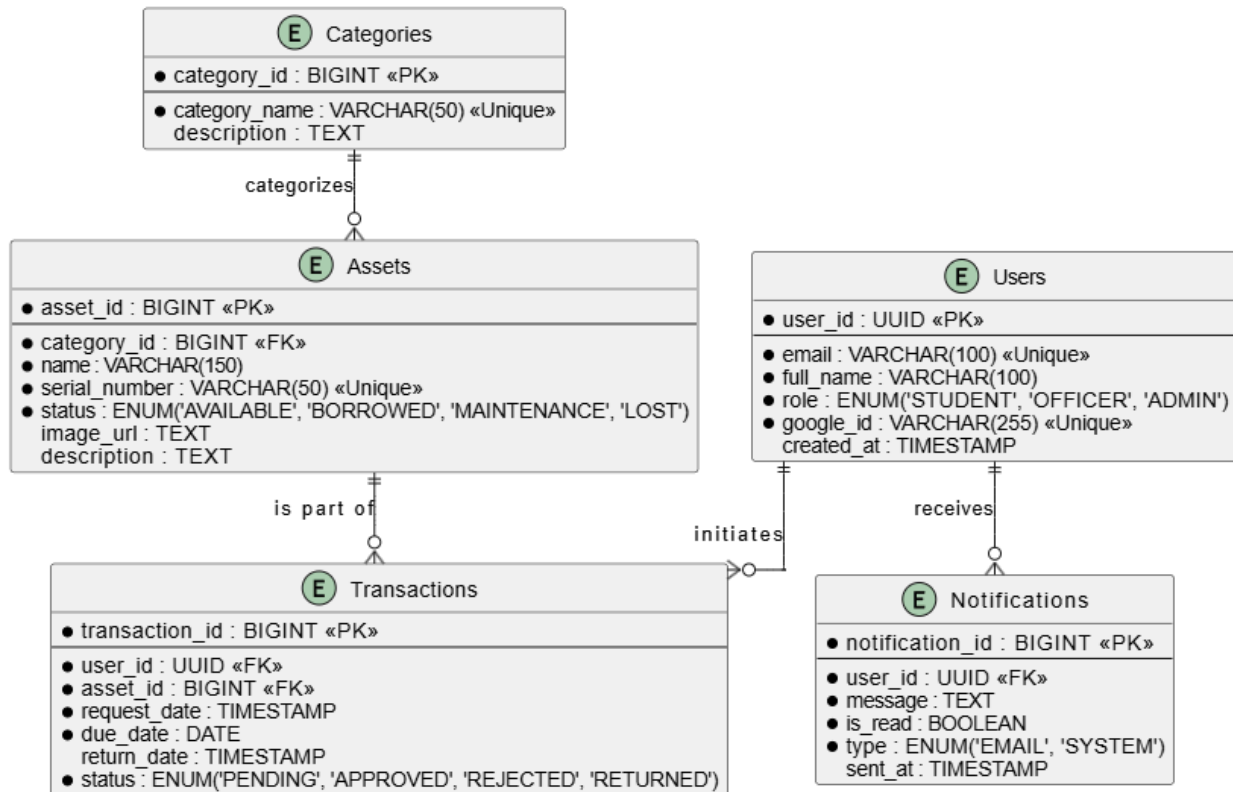
Common Error Codes

- AUTH-001: Invalid credentials / Google OAuth failed
- AUTH-002: Token expired
- AUTH-003: Insufficient permissions (Role-based restriction)
- VALID-001: Validation failed (Input data does not meet requirements)
- DB-001: Resource not found
- DB-002: Duplicate entry (e.g., duplicate Serial Number)
- BUSINESS-001: Asset unavailable (Item is already borrowed)
- SYSTEM-001: Internal server error

6.0 DATABASE DESIGN

6.1 Entity Relationship Diagram

Entity Relationship Diagram for AssetFlow



Detailed Relationships:

- **One-to-Many:** Category → Assets (One category can contain many different items)
- **One-to-Many:** User → Transactions (One user can have multiple borrowing records over time)
- **One-to-Many:** Asset → Transactions (One asset can have a history of being borrowed many times)
- **One-to-Many:** User → Notifications (One user can receive multiple system alerts)

Key Tables:

1. **users** – User accounts, Google OAuth identity, and role-based permissions
2. **assets** – Detailed inventory catalog, serial numbers, and current availability
3. **categories** – Grouping for assets to facilitate organized filtering and reporting
4. **transactions** – Lifecycle tracking for borrowing requests, approvals, and returns

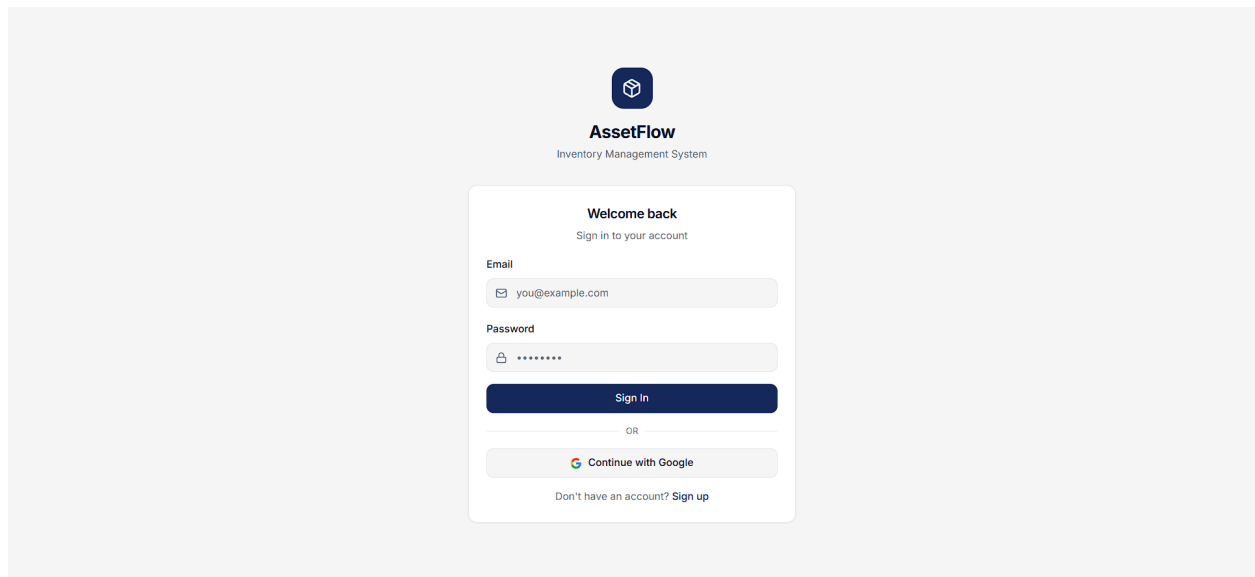
5. **notifications** – Log of system alerts and status updates for students and officers

Table Structure Summary:

- **users:** id, email, full_name, role, google_id, created_at
- **assets:** id, category_id, name, serial_number, status, image_url
- **categories:** id, name, description
- **transactions:** id, user_id, asset_id, request_date, due_date, return_date, status
- **notifications:** id, user_id, message, is_read, sent_at

7.0 UI/UX DESIGN

7.1 Web Application Wireframes





Create Account

Join AssetFlow today

First Name

Last Name

Maria

Santos

Email

you@example.com

Password

Confirm Password

Create Account

Already have an account? [Sign in](#)

**AssetFlow**
INVENTORY SYSTEM

Dashboard

My Transactions

DEMO: SWITCH ROLE

Student

MS Maria Santos

Inventory Dashboard

Welcome back, Maria. Browse available assets.

6
Total Assets

3
Available

2
Borrowed

1
Maintenance

Search assets by name, serial number, or category...

All

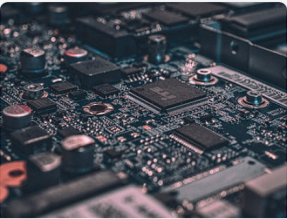
Available

Borrowed

Maintenance



Digital Oscilloscope
OSC-2024-001
Electronics
Available



3D Printer - Prusa MK4
PRT-2024-003
Tools
Borrowed



Arduino Mega Kit
ARD-2024-012
Electronics
Available







**AssetFlow**
INVENTORY SYSTEM

Dashboard

My Transactions

DEMO: SWITCH ROLE
Student
MS Maria Santos Student

← Back



Available

Digital Oscilloscope

Tektronix TBS1052B 50MHz 2-channel digital oscilloscope for circuit analysis.

Serial NumberOSC-2024-001

CategoryElectronics

Added2024-01-15

Submit Borrow Request

Request submitted!
Your borrow request for Digital Oscilloscope has been sent.

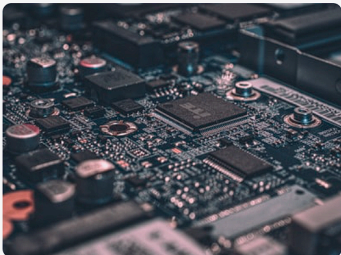
**AssetFlow**
INVENTORY SYSTEM

Dashboard

My Transactions

DEMO: SWITCH ROLE
Student
MS Maria Santos Student

← Back



Borrowed

3D Printer - Prusa MK4

High-quality FDM 3D printer for prototyping and project models.

Serial NumberPRT-2024-003

CategoryTools

Added2024-02-10

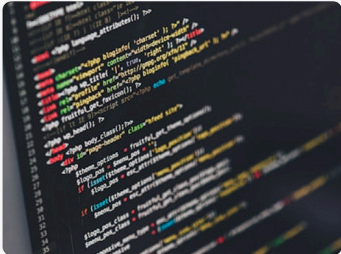
 **AssetFlow**
INVENTORY SYSTEM

Dashboard

My Transactions

DEMO: SWITCH ROLE
Student
MS Maria Santos Student

← Back



Maintenance

Projector - Epson EB-X51
XGA projector with 3800 lumens, ideal for presentations and lab demos.

Serial Number

PRJ-2024-005

Category

Audio/Visual

Added

2024-01-20

 **AssetFlow**
INVENTORY SYSTEM

Dashboard

My Transactions

DEMO: SWITCH ROLE
Student
MS Maria Santos Student

My Transactions
Track your borrow requests and return history.

Digital Oscilloscope

Requested: 2026-02-28 Due: 2026-03-14

Pending

3D Printer - Prusa MK4

Requested: 2026-02-10 Due: 2026-03-10

Approved

AssetFlow
INVENTORY SYSTEM

Dashboard

Borrow Requests

Add Asset

DEMO: SWITCH ROLE

Admin

Admin Officer

Inventory Dashboard

Welcome back, Admin. Manage your organization's assets.

6
Total Assets

3
Available

2
Borrowed

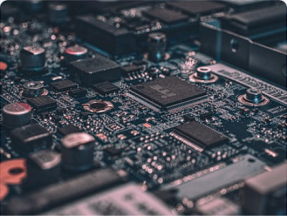
1
Maintenance

Search assets by name, serial number, or category...

AllAvailableBorrowedMaintenance



Digital Oscilloscope
OSC-2024-001
Electronics
Available



3D Printer - Prusa MK4
PRT-2024-003
Tools
Borrowed



Arduino Mega Kit
ARB-2024-012
Electronics
Available



AssetFlow - Inventory Management System - Google Chrome

AssetFlow
INVENTORY SYSTEM

Dashboard

Borrow Requests

Add Asset

DEMO: SWITCH ROLE

Admin

Admin Officer

Borrow Requests

Review and manage student borrow requests.

Student	Asset	Request Date	Due Date	Status	Actions
Maria Santos	Digital Oscilloscope	2026-02-28	2026-03-14	Pending	ApproveReject
Juan Dela Cruz	Arduino Mega Kit	2026-02-25	2026-03-11	Pending	ApproveReject
Maria Santos	3D Printer - Prusa MK4	2026-02-10	2026-03-10	Approved	Mark Returned
Juan Dela Cruz	MacBook Pro 14"	2026-01-15	2026-02-15	Returned	

AssetFlow
INVENTORY SYSTEM

Dashboard
Borrow Requests
Add Asset

DEMO: SWITCH ROLE
Admin

AO Admin Officer

Add New Asset

Add a new item to the organization's inventory.

Asset Name
e.g., Digital Oscilloscope

Serial Number
e.g., OSC-2024-001

Category
Select category

Status
Available

Description
Brief description of the asset...

Upload File (Image / PDF)
Click to upload or drag and drop
PNG, JPG, PDF up to 10MB

Save Asset **Cancel**

Header

[AssetFlow Logo] [Search Bar] [Filter Dropdown: Status] [Notifications Icon] [User Menu Dropdown]

- Search Bar: Search by asset name or serial number
- Notifications Icon: Shows unread system alerts
- User Menu: Profile, Logout

Content

Each Asset Card contains:

- Asset Image
- Asset Name
- Serial Number
- Category Tag
- Status Badge:
 - Green – Available
 - Red – Borrowed
 - Yellow – Maintenance

Asset Detail Page

Back Button

Large Asset Image

Asset Information Section:

- Name
- Serial Number
- Category
- Current Status
- Description (if available)

Action Buttons (Role-Based):

- Student Role:
 - "Request to Borrow" button (visible only if status = Available)
- Admin/Laboratory Keeper:
 - Edit Asset
 - Delete Asset
 - Mark as Under Maintenance

Additional Section:

Transaction History Preview (last 3 borrow records)

Borrow Request Page (Student)

Asset Summary Card

Expected Return Date Picker

Submit Borrow Request button

System Feedback:

- Success notification banner
- Validation error messages

My Transactions Page (Student)

Page Title: My Borrowed Assets

Table/List View:

- Asset Name
- Borrow Date
- Due Date

- Status (Pending, Approved, Rejected, Returned)
- View Details button

Status indicators are color-coded.

Admin Dashboard

Sidebar Navigation:

- Dashboard
- Assets
- Categories
- Borrow Requests
- Transactions History
- Users
- Logout

Asset Management (Admin)

Add New Asset button

Asset Table:

- Image Thumbnail
- Name
- Serial Number
- Category
- Status
- Edit button
- Delete button

Search and filter options available.

Borrow Request Management (Admin)

Request List:

- Student Name
- Asset
- Request Date
- Expected Return Date

- Status

Action Buttons:

- Approve
- Reject
- Mark as Returned

7.2 Mobile Application Wireframes



AssetFlow

Inventory Management System

Welcome back

Sign in to your account

Email

 you@example.com

Password



Sign In

OR

 Continue with Google

Don't have an account? [Sign up](#)



Create Account

Join AssetFlow today

First Name

Maria

Last Name

Santos

Email

 you@example.com

Password



Confirm Password



Create Account

Already have an account? [Sign in](#)

 **AssetFlow**
INVENTORY SYSTEM

Dashboard

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DEMO: SWITCH ROLE

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MS

Maria Santos

Student

Inventory Dashboard

Welcome back, Maria. Browse available assets.

6

Total Assets

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Available

2

Borrowed

1

Maintenance

Q

All

Available

Borrowed

Maintenance

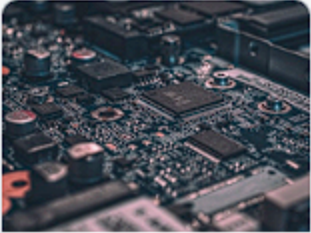


Digital Oscilloscope

OSC-2024-001

Electronics

Available



3D Printer - Prusa MK4

PRT-2024-003

Tools

Borrowed



Arduino Mega Kit

ARD-2024-012

Electronics

Available



Projector - Epson E...

PRJ-2024-005

Audio/Visual

Maintenance



28

**AssetFlow**
INVENTORY SYSTEM

 Dashboard

 My Transactions

DEMO: SWITCH ROLE

Student

MS

Maria Santos

 Student

[→]

[← Back](#)



Available

Digital Oscilloscope

Tektronix TBS1052B 50MHz 2-channel digital oscilloscope for circuit analysis.

#	Serial Number	OSC-2024-001
	Category	Electronics
	Added	2024-01-15

Submit Borrow Request

**AssetFlow**
INVENTORY SYSTEM

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My Transactions

DEMO: SWITCH ROLE

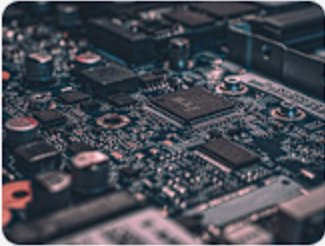
Student

MS

Maria Santos

Student

← Back



Borrowed

3D Printer - Prusa MK4

High-quality FDM 3D printer for prototyping and project models.

#	Serial Number	PRT-2024-003
	Category	Tools
	Added	2024-02-10

**AssetFlow**
INVENTORY SYSTEM

Dashboard

My Transactions

DEMO: SWITCH ROLE

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MS

Maria Santos

Student

← Back



Maintenance

Projector - Epson EB-X51

XGA projector with 3800 lumens, ideal for presentations and lab demos.

#	Serial Number	PRJ-2024-005
	Category	Audio/Visual
	Added	2024-01-20

**AssetFlow**
INVENTORY SYSTEM

Dashboard

My Transactions

DEMO: SWITCH ROLE

Student

MS **Maria Santos**
Student

My Transactions

Track your borrow requests and return history.



Digital Oscilloscope
Requested: 2026-02-28 Due: 2026-03-14 Pending



3D Printer - Prusa MK4
Requested: 2026-02-10 Due: 2026-03-10 Approved

 **AssetFlow**
INVENTORY SYSTEM

Dashboard

Borrow Requests

Add Asset

DEMO: SWITCH ROLE

Admin

Admin Officer

Inventory Dashboard

Welcome back, Admin. Manage your organization's assets.

6

Total Assets

3

Available

2

Borrowed

1

Maintenance

Q

All

Available

Borrowed

Maintenance



Digital Oscilloscope

Available

DSC-2024-001

Electronics



3D Printer - Prusa MK4

Borrowed

PRT-2024-003

Tools



Arduino Mega Kit

Available

ARD-2024-012

Electronics



Projector - Epson E...

Maintenance

PRJ-2024-005

Audio/Visual





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 **AssetFlow**
INVENTORY SYSTEM

Dashboard

Borrow Requests

Add Asset

DEMO: SWITCH ROLE

Admin

AO

Admin Officer

Admin

Borrow Requests

Review and manage student borrow requests.

Student	Asset	Request Date	Due Date	Status
Maria Santos	Digital Oscilloscope	2026-02-28	2026-03-14	Pending
Juan Dela Cruz	Arduino Mega Kit	2026-02-25	2026-03-11	Pending
Maria Santos	3D Printer - Prusa MK4	2026-02-10	2026-03-10	Approved
Juan Dela Cruz	MacBook Pro 14"	2026-01-15	2026-02-15	Returned

 **AssetFlow**
INVENTORY SYSTEM

Dashboard

Borrow Requests

+

Add Asset

DEMO: SWITCH ROLE

Admin

AO

Admin Officer

Admin

Add New Asset

Add a new item to the organization's inventory.

Asset Name

e.g., Digital Oscilloscope

Serial Number

e.g., OSC-2024-001

Category

Select category

Status

Available

Description

Brief description of the asset...

Upload File (Image / PDF)



Click to upload or drag and drop
PNG, JPG, PDF up to 10MB

Save Asset

Cancel

Bottom Navigation

[ Home] [ My Borrows] [ Notifications] [ Profile]

Navigation adjusts based on user role.

Login Screen

Email Field

Password Field

Login Button

Google Login Button

Error message display area

Loading spinner during authentication

Home Screen (Inventory)

Search Bar

RecyclerView List (2-column grid)

Each Asset Card:

- Image
-
- Name
- Status badge
- Tap to view details

Mobile Features:

- Pull-to-refresh
- Smooth scrolling
- Offline image caching

Asset Detail Screen

Back Arrow

Large ImageView

Asset Details:

- Name
- Serial Number
- Category
- Status

Action Button:

- Student Role: Borrow button (if Available)
- Admin Role: Approve/Reject controls

Confirmation dialog before submitting request.

My Borrows Screen

RecyclerView List:

- Asset Name
- Borrow Date
- Due Date
- Status Label

Swipe gesture:

- Swipe left to view transaction details

Admin Mobile Screen (Role-Based)

Request Approval List:

- Student Name
- Asset
- Status

Buttons:

- Approve
- Reject
- Mark Returned

Mobile-Specific Features:

- Touch-optimized buttons (minimum 44x44px)
- Pull-to-refresh gesture
- RecyclerView with smooth animations
- JWT stored securely (Encrypted SharedPreferences)
- Error handling with Snackbar alerts
- Bottom navigation for core actions

Design System:

- **Colors:** Primary – #2563EB (Blue), Secondary – #7C3AED (Purple), Success – #10B981 (Green), Warning – #F59E0B (Amber), Error – #EF4444 (Red)
- **Typography:** Inter font family
- **Spacing:** 8px grid system, consistent padding and margins
- **Components:**
 - Reusable Card Component (AssetCard)
 - Status Badge Component
 - Modal Confirmation Dialog
 - Toast / Snackbar Notification
 - Form Input with validation state
- **Responsive:**
 - Mobile: < 640px
 - Tablet: 768px
 - Desktop: ≥ 1024px
 - Mobile-first approach for web responsiveness.

8.0 PLAN

8.1 Project Timeline

Phase 1: Planning & Design (Week 1-2)

Week 1: Requirements & Architecture

Day 1-2: Project setup and documentation for AssetFlow

Day 3-4: Complete Functional and Non-Functional Requirements

Day 5-7: System architecture design (C4/Component Diagrams)

Week 2: Detailed Design

Day 1-2: API specification (Endpoints and JWT Auth)

Day 3-4: Database design (ERD and Table Summary)

Day 5-6: UI/UX wireframes (Figma for Web and Mobile)

Day 7: Implementation plan finalization

Phase 2: Backend Development (Week 3-4)

Week 3: Foundation

Day 1: Spring Boot setup with required dependencies

Day 2: Database configuration and JPA entities (Assets, Users)

Day 3: JWT authentication and Google OAuth implementation

Day 4: User management and role-based endpoints

Day 5: Asset CRUD operations

Week 4: Core Features

Day 1: Transaction/Borrowing logic functionality

Day 2: Category management and filtering logic

Day 3: Search functionality for inventory

Day 4: Error handling and data validation

Day 5: API documentation and Postman testing

Phase 3: Web Application (Week 5-6)

Week 5: Frontend Foundation

Day 1: React setup with TypeScript and Tailwind CSS

Day 2: Authentication pages (Google Login)

Day 3: Inventory listing page for Officers

Day 4: Asset detail and management views

Day 5: Borrowing request approval interface

Week 6: Complete Web Features

Day 1: Transaction history tracking

Day 2: Notification center for status updates

Day 3: Admin dashboard and report generation

Day 4: Responsive design polish for desktop/tablet

Day 5: API integration and unit testing

Phase 4: Mobile Application (Week 7-8)

Week 7: Android Foundation

Day 1: Android Studio setup and Kotlin/XML project structure

Day 2: Mobile authentication screens

Day 3: Asset browsing and category filtering

Day 4: Borrowing request form for Students

Day 5: Retrofit API service layer integration

Week 8: Complete Mobile App

Day 1: "My Borrows" personal tracking view

Day 2: Push notification/Alert system

Day 3: UI polish and smooth XML transitions

Day 4: Testing on Android Emulator/Physical Device

Day 5: APK generation and mobile documentation

Phase 5: Integration & Deployment (Week 9-10)

Week 9: Integration Testing

Day 1: End-to-end testing across Web, Mobile, and Backend

Day 2: Bug fixes and performance optimization

Day 3: Security review (JWT and OAuth validation)

Day 4: Load testing for concurrent users

Day 5: Documentation final updates

Week 10: Deployment

Day 1: Backend deployment (Render/Railway)

Day 2: Web app deployment (Vercel/Netlify)

Day 3: Mobile APK distribution

Day 4: Final verification and testing

Day 5: Final project submission

Milestones:

- **M1 (End Week 2):** Final SDD with ERD, Component Diagrams, and Wireframes
- **M2 (End Week 4):** Backend API Ready Functional REST API with Google OAuth and Asset CRUD
- **M3 (End Week 6):** Functional React Dashboard for Officer asset management.
- **M4 (End Week 8):** Functional Android App for Student borrowing requests
- **M5 (End Week 10):** Fully integrated system deployed on cloud infrastructure

Critical Path:

1. Authentication system (Week 3)

2. Asset inventory API (Week 3-4)
3. Borrowing request functionality (Week 4)
4. Approval/Return process (Week 6)
5. Cross-platform testing (Week 9)

Risk Mitigation:

- Start with simplest working version of each feature
- Test integration points early and often
- Keep backup of working versions
- Focus on core functionality before enhancements